SCEC TPV104, strike slip rate at the nine on-fault stations (rows: depth 3 / 7.5 / 12 km; columns: strike)

x2_-12_x3_3 (strike -12 km, depth 3 km)

x2_0_x3_3 (strike 0 km, depth 3 km)

x2_12_x3_3 (strike 12 km, depth 3 km)

x2_-9_x3_7.5 (strike -9 km, depth 7.5 km)

x2_0_x3_7.5 (strike 0 km, depth 7.5 km)

x2_9_x3_7.5 (strike 9 km, depth 7.5 km)

x2_-12_x3_12 (strike -12 km, depth 12 km)

x2_0_x3_12 (strike 0 km, depth 12 km)

x2_12_x3_12 (strike 12 km, depth 12 km)

strike slip rate (m/s)

time (s)

DRDG3D 200 m O4

SeisSol O5

MFEM mixed-flux RK45 p=2 (gold)

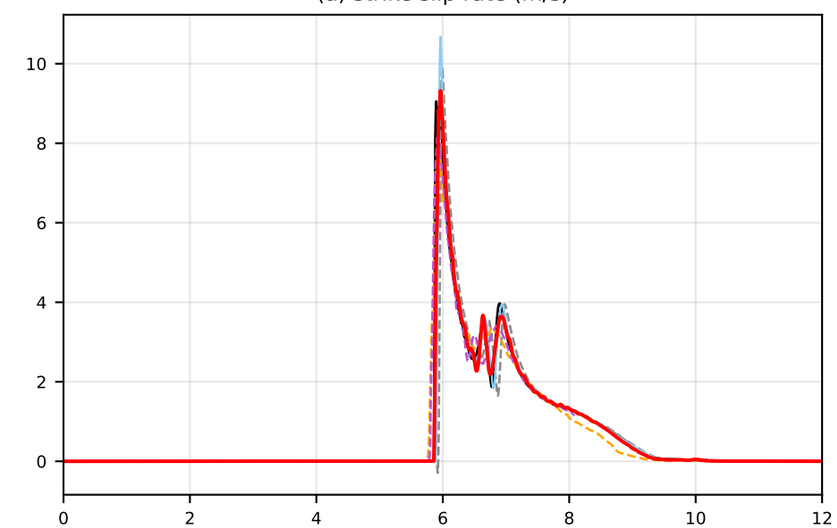
Nitsche p=1 200 m (Vista 1075919)

Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)

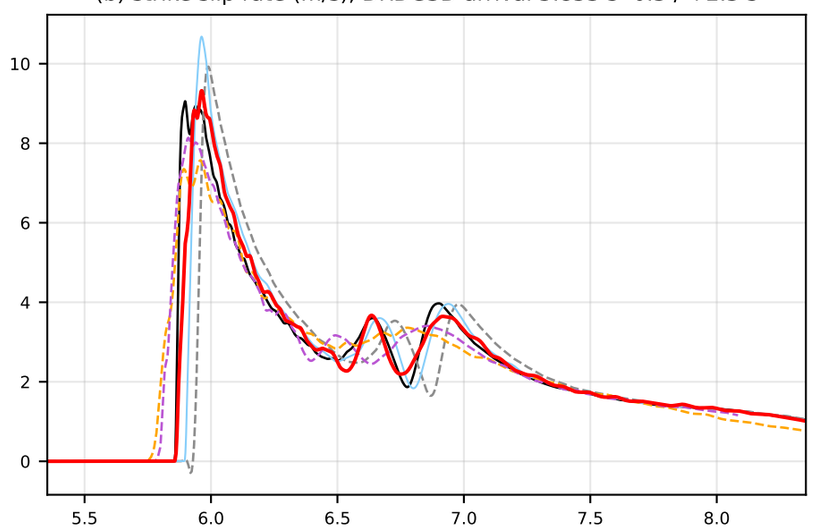
Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_-12_x3_3 (strike -12 km, depth 3 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt +0.003 s slip -1.45% peak +2.6%

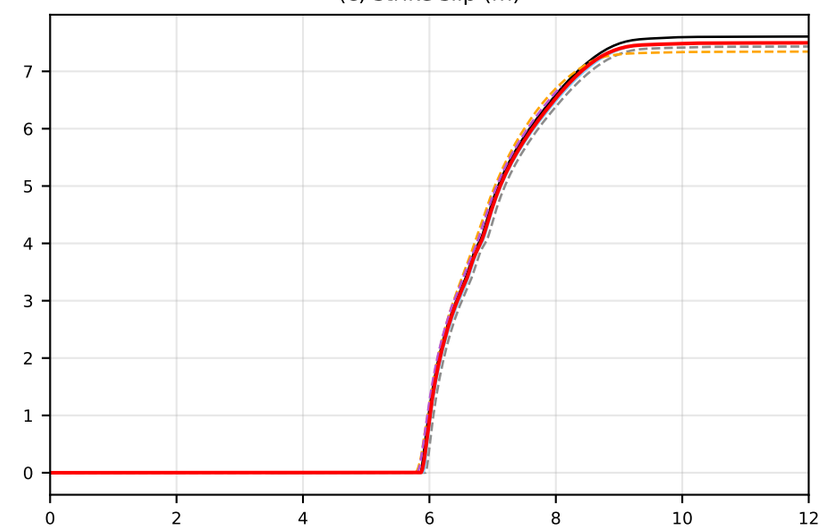
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.853 s -0.5 / +2.5 s

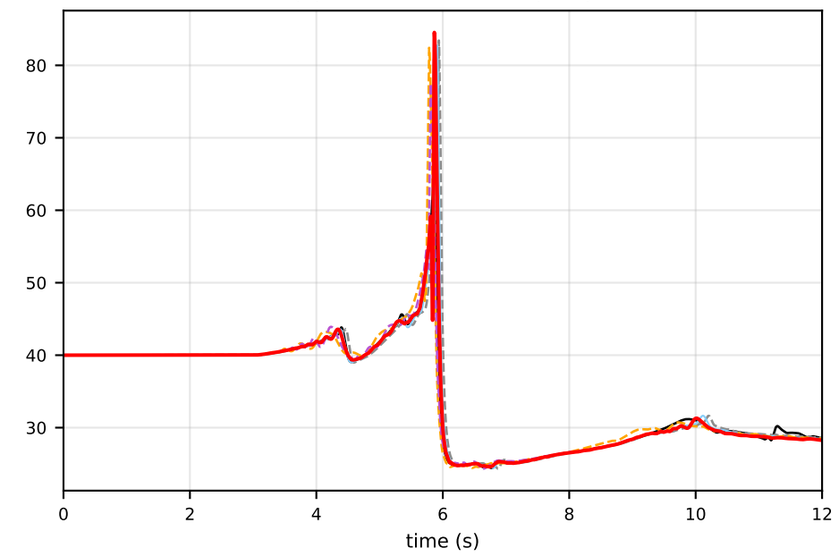


(c) strike slip (m)

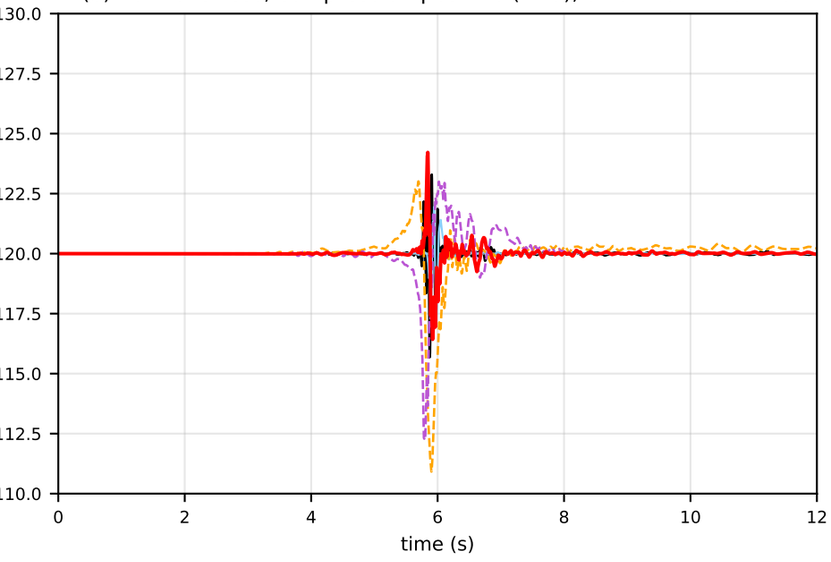


time (s)

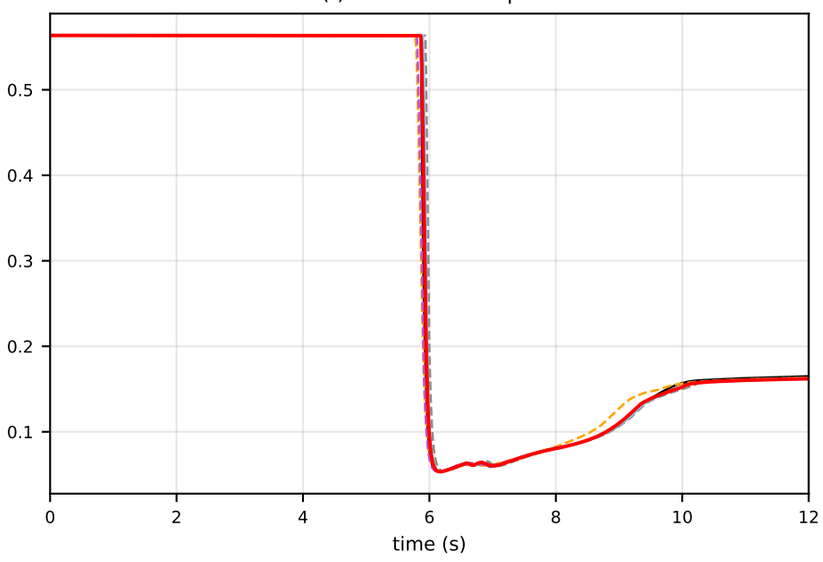
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa



(f) state variable psi



DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_0_x3_3 (strike 0 km, depth 3 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt -0.008 s slip -0.13% peak +4.8%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 2.620 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

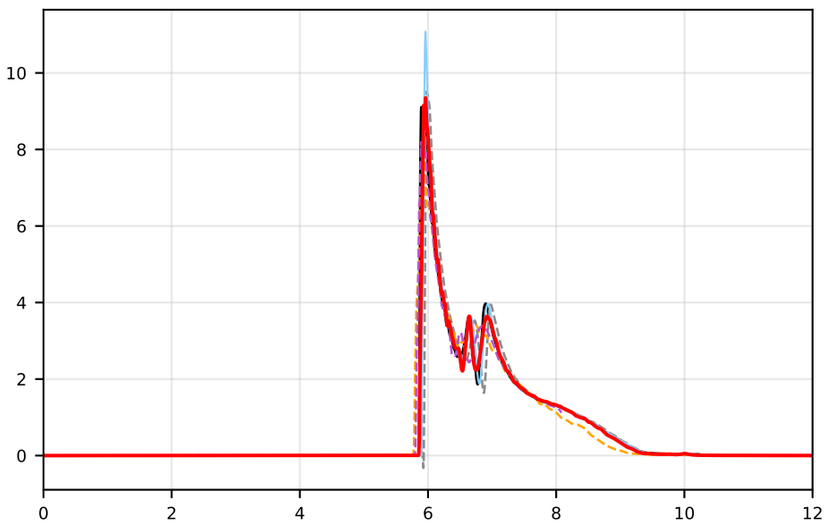
(f) state variable psi

time (s)

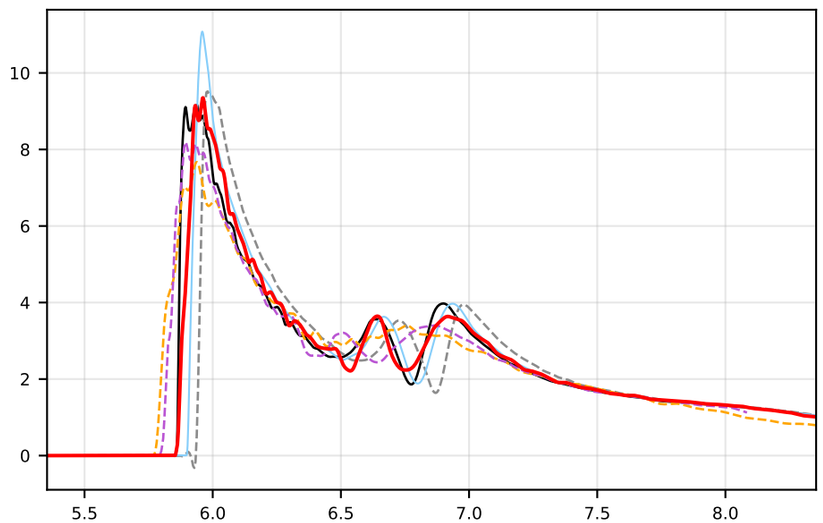
DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_12_x3_3 (strike 12 km, depth 3 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt -0.001 s slip -1.44% peak +2.0%

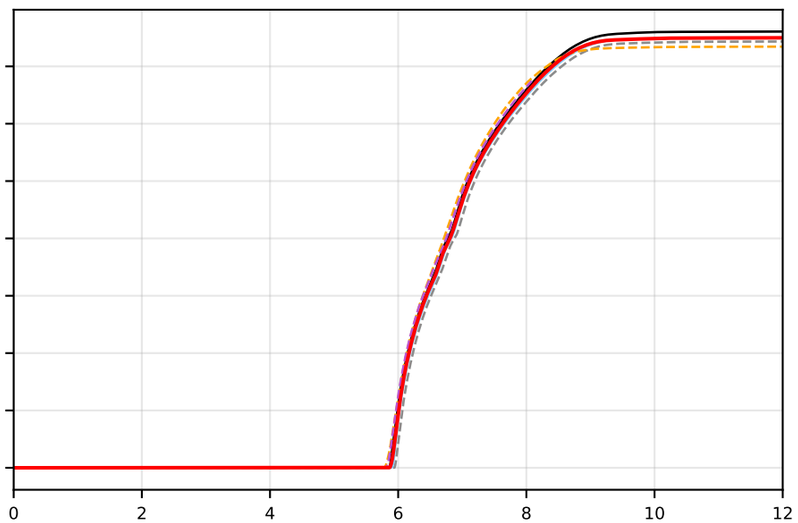
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.854 s -0.5 / +2.5 s

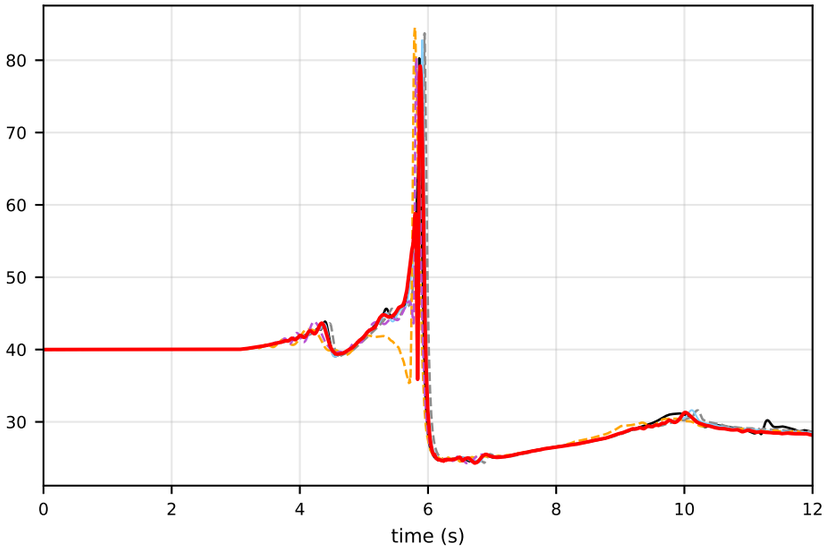


(c) strike slip (m)

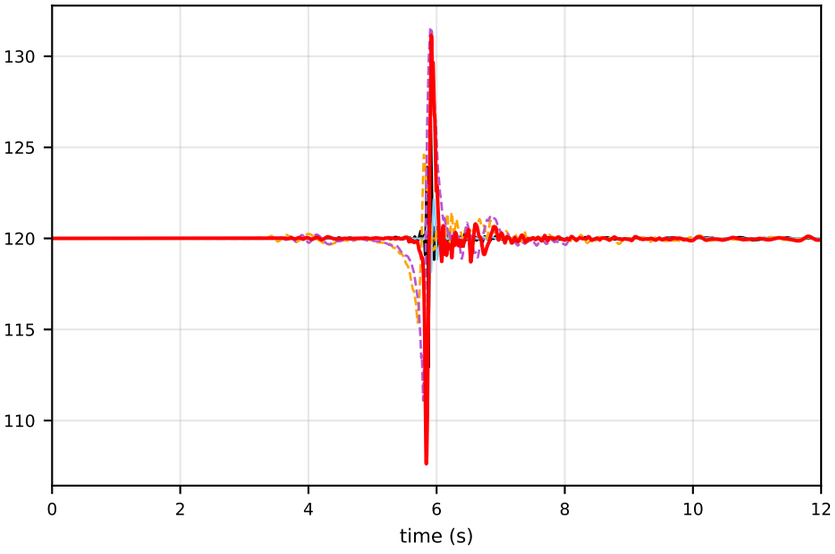


time (s)

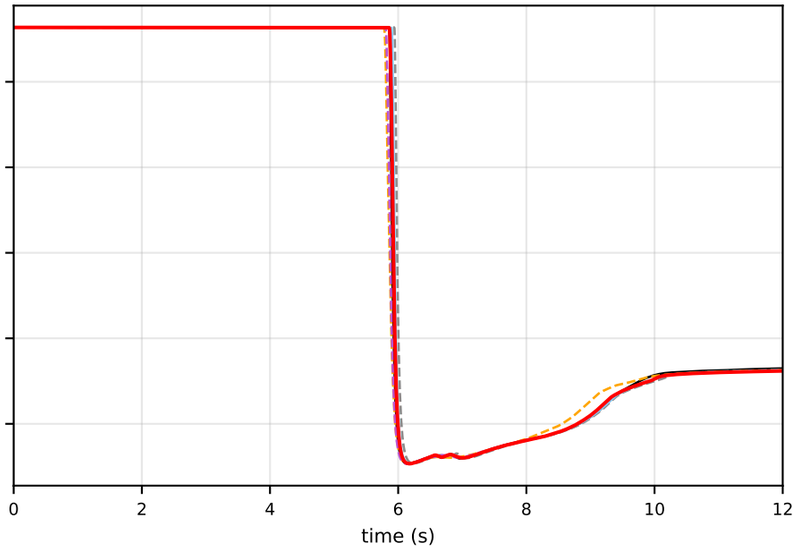
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); autoscaled: data span 107.6-131.6 MPa

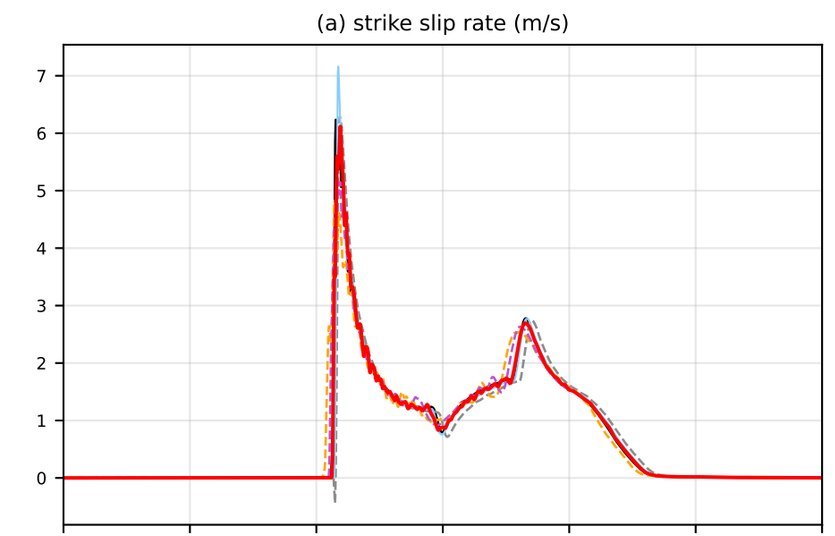


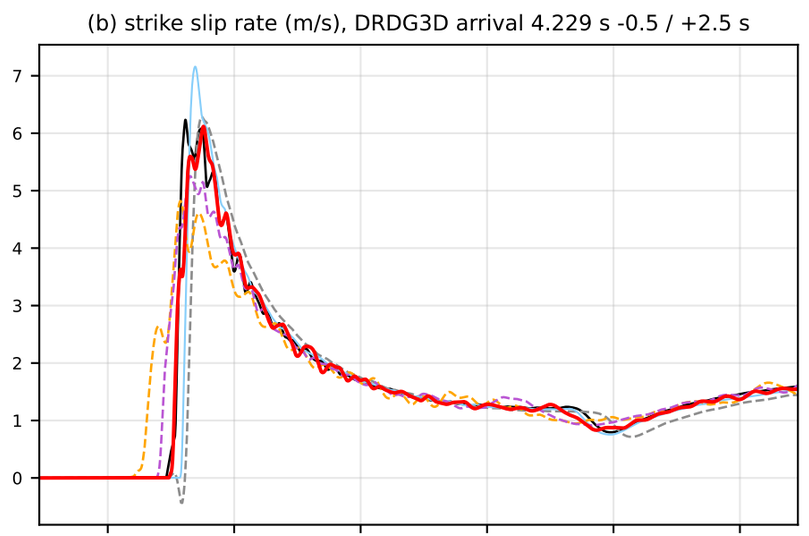
(f) state variable psi



DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)

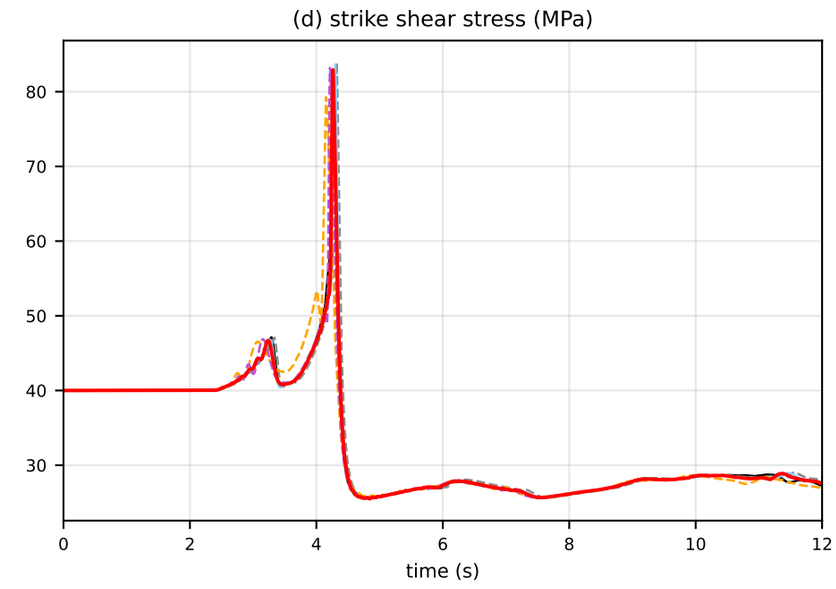
SCEC TPV104 station x2_-9_x3_7.5 (strike -9 km, depth 7.5 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt +0.011 s slip -0.49% peak -1.8%

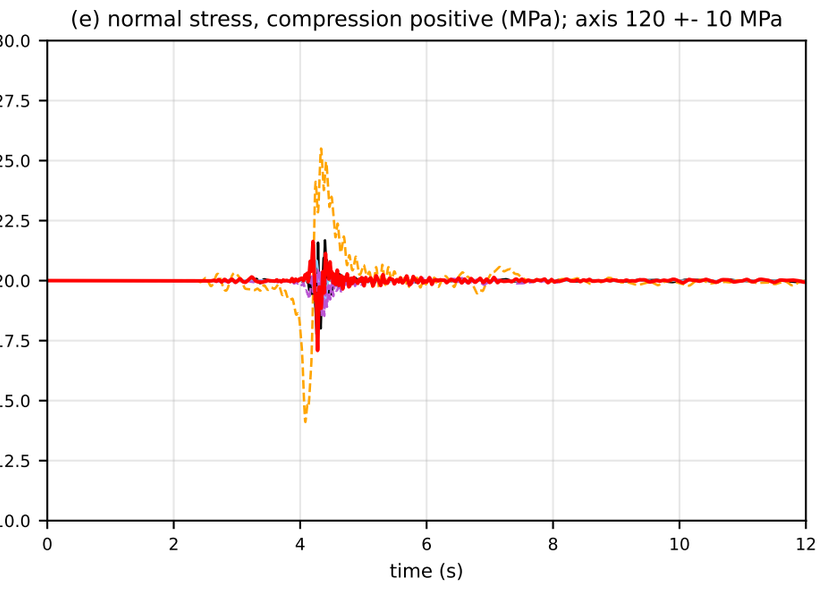


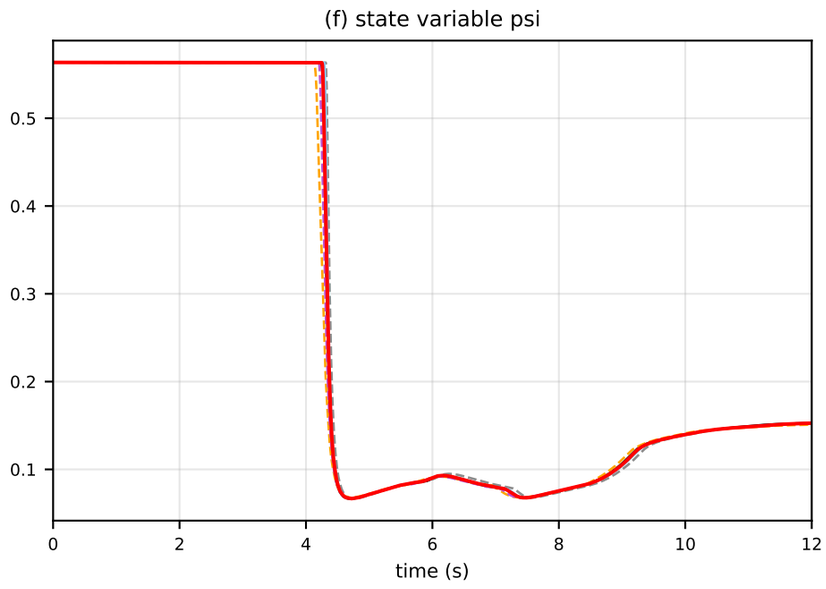




time (s)







DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_0_x3_7.5 (strike 0 km, depth 7.5 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt -0.003 s slip -0.09% peak +0.2%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 0.574 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

(f) state variable psi

time (s)

DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_9_x3_7.5 (strike 9 km, depth 7.5 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt +0.012 s slip -0.48% peak   -7.7%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 4.229 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); autoscaled: data span 111.3-130.6 MPa

(f) state variable psi

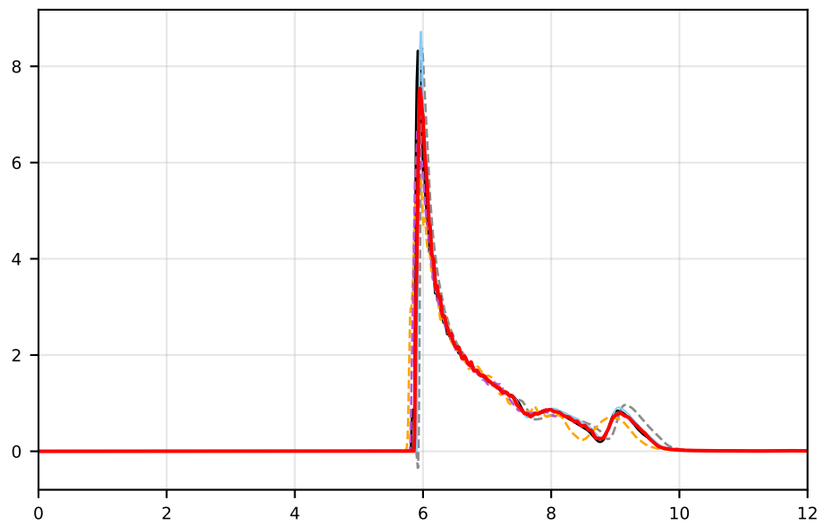
time (s)

time (s)

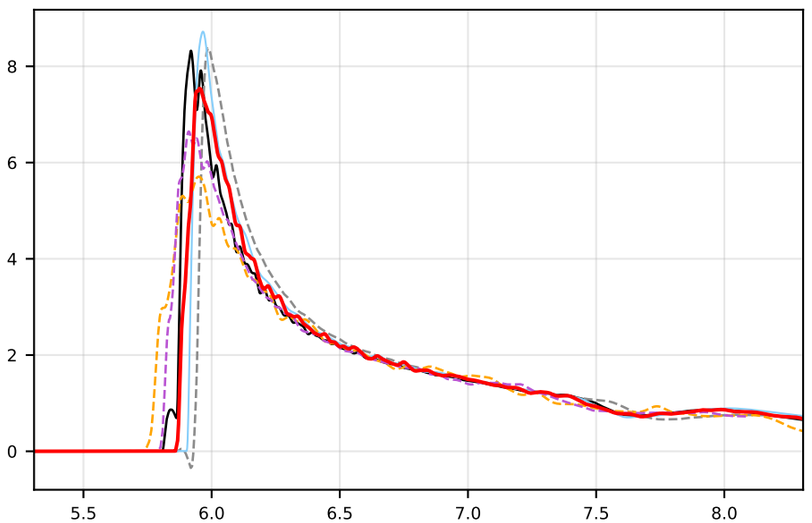
DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_-12_x3_12 (strike -12 km, depth 12 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt +0.050 s slip -0.97% peak -9.5%

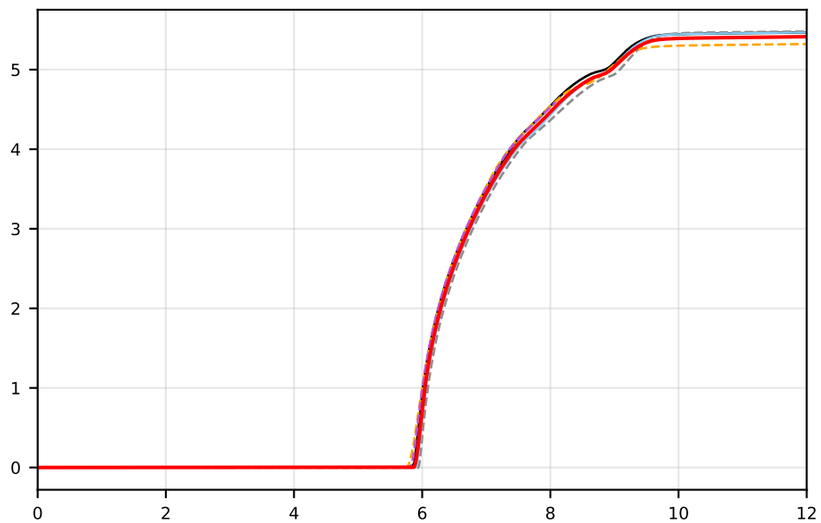
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.807 s -0.5 / +2.5 s

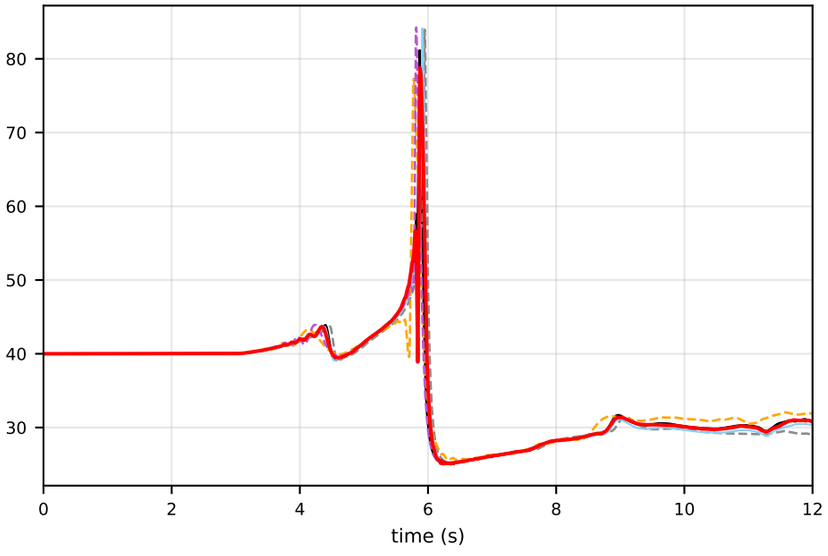


(c) strike slip (m)

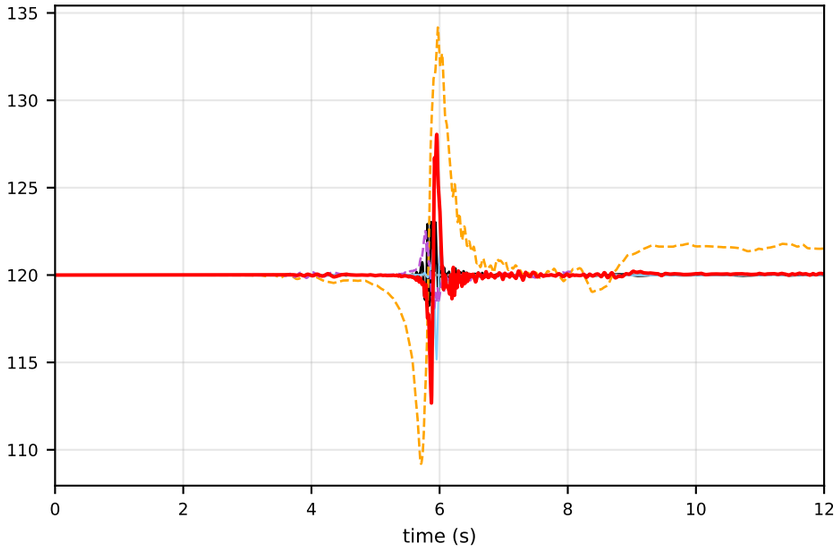


time (s)

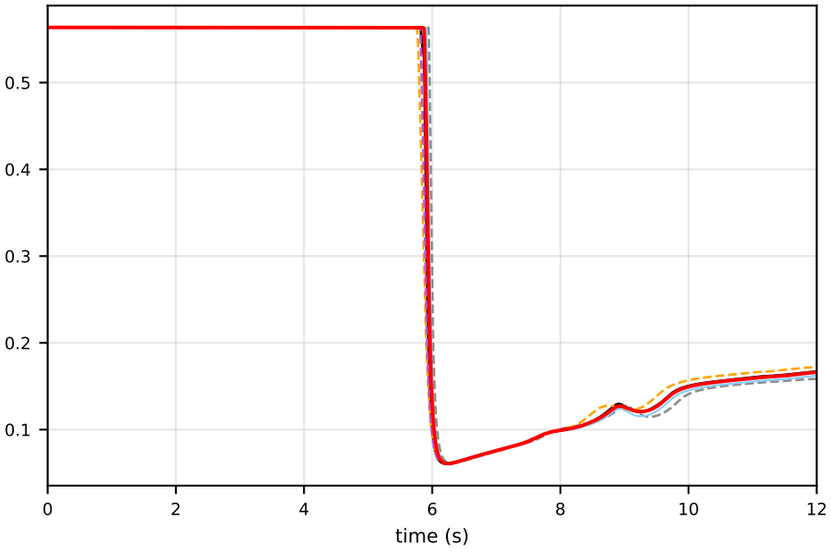
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); autoscaled: data span 109.2-134.2 MPa



(f) state variable psi



DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_0_x3_12 (strike 0 km, depth 12 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt +0.045 s slip +0.10% peak +12.3%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 2.552 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

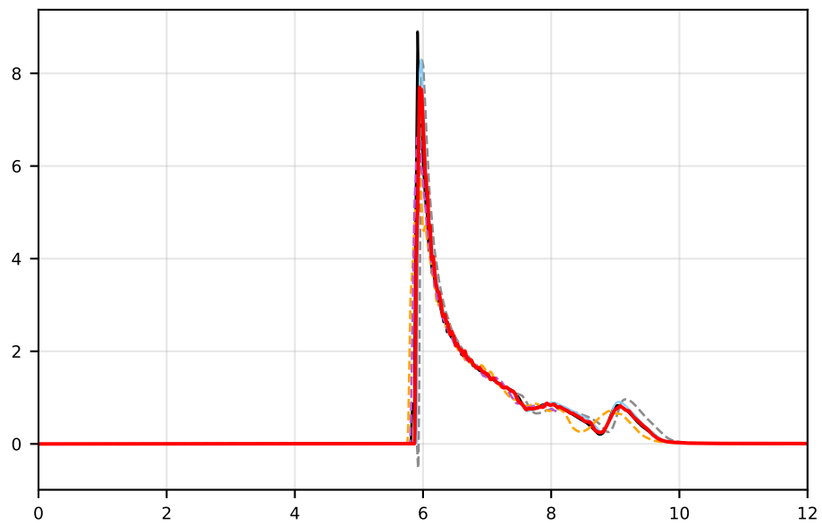
(f) state variable psi

time (s)

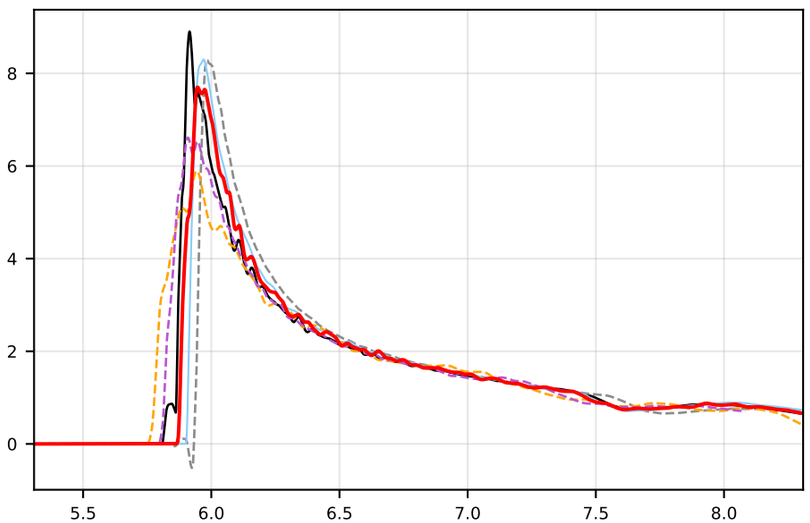
DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)

SCEC TPV104 station x2_12_x3_12 (strike 12 km, depth 12 km)
Nitsche p=2 200 m (Vista 1075917) vs DRDG3D: dt +0.056 s slip -0.94% peak -13.5%

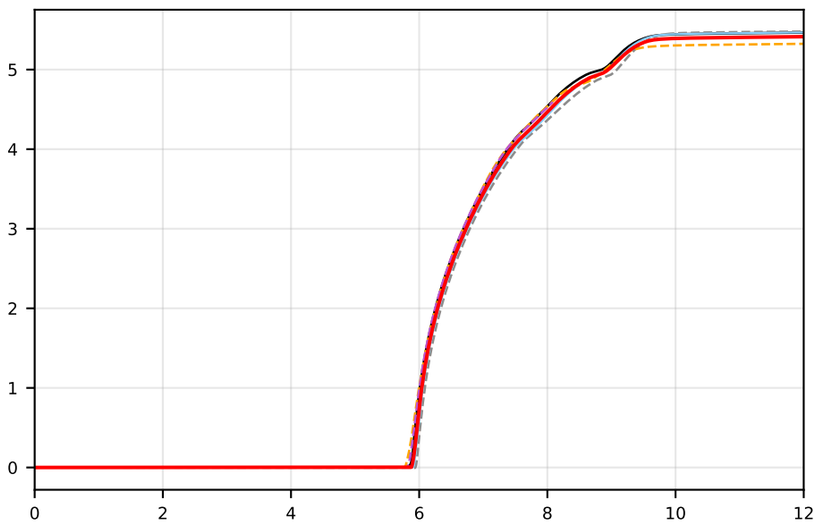
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.809 s -0.5 / +2.5 s

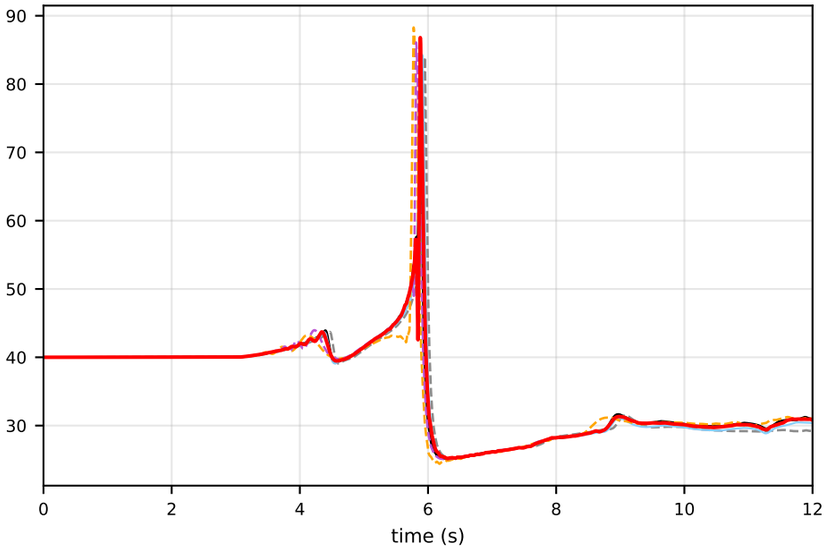


(c) strike slip (m)

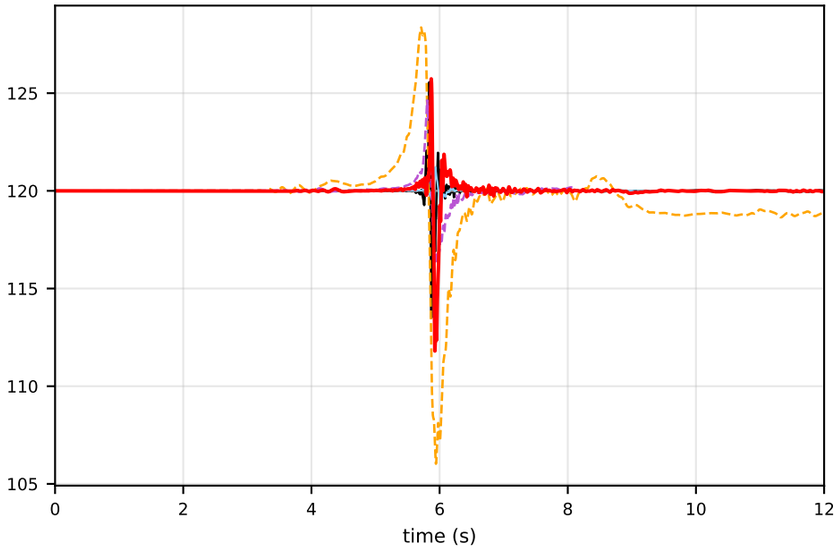


time (s)

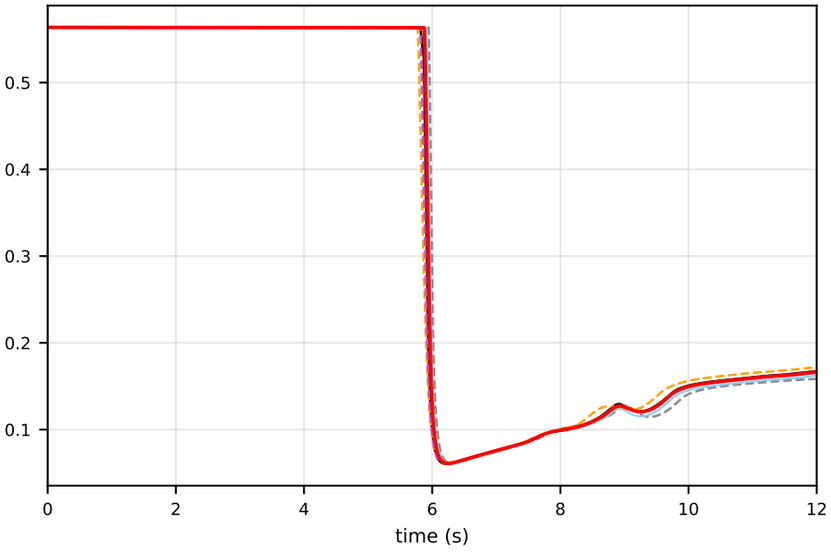
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); autoscaled: data span 106.0-128.4 MPa



(f) state variable psi



DRDG3D 200 m O4 | SeisSol O5 | MFEM mixed-flux RK45 p=2 (gold) | Nitsche p=1 200 m (Vista 1075919) | Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s) | Nitsche p=2 200 m (Vista 1075917)